My reproducible manuscript

Roos Mulder

```
# Load penguins
library(palmerpenguins)
```

```
Warning: package 'palmerpenguins' was built under R version 4.4.3
```

```
library(ggplot2)
```

```
Warning: package 'ggplot2' was built under R version 4.4.3
```

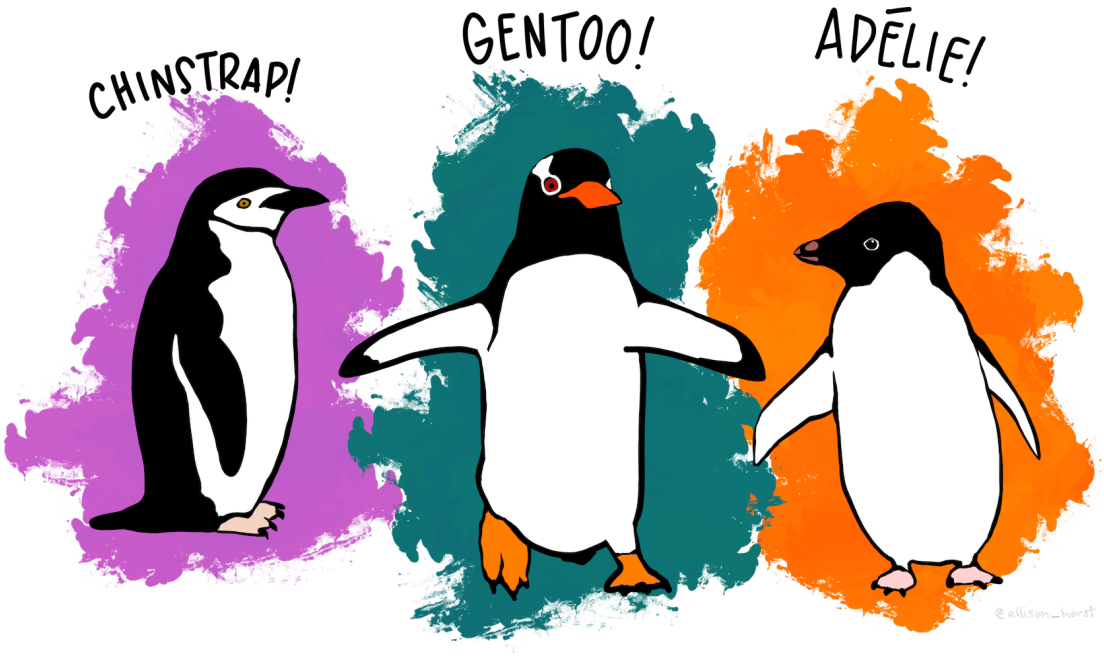
```
# Read data
penguins = penguins
```

Meet Quarto

Quarto enables you to weave together content and executable code into a finished document. To learn more about Quarto see https://quarto.org.

Meet the penguins

The `penguins` data from the palmerpenguins (Horst, Hill, and Gorman (2022)) package contains size measurements for 344 penguins from three species observed on three islands in the Palmer Archipelago, Antarctica.



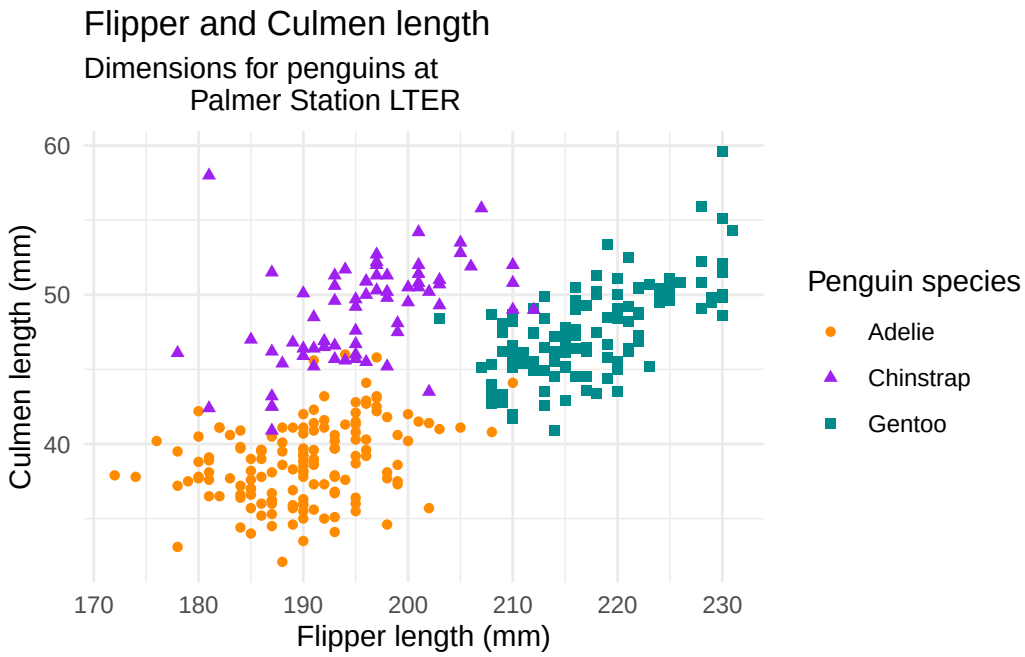
Want to know more about how we're destroying our planet and everything that lives in it (including penguins)? Check out: Forcada and Trathan (2009)

Results

```
# Plot penguins
ggplot(penguins,
       aes(x = flipper_length_mm, y = bill_length_mm)) +
        geom_point(aes(color = species, shape = species)) +
        scale_color_manual(values = c("darkorange","purple","cyan4")) +
        labs(title = "Flipper and Culmen length",
             subtitle = "Dimensions for penguins at
             Palmer Station LTER",
             x = "Flipper length (mm)",
             y = "Culmen length (mm)",
             color = "Penguin species",
             shape = "Penguin species") +
        theme_minimal()
```

```
Warning: Removed 2 rows containing missing values or values outside the scale range
```

(`geom_point()`).



A simple calculation might look like this 66

There are 68 Chinstrap penguins

There are 124 Gentoo penguins

The average bill depth is 17.151 millimeters

A function

```
do_addition <- function(number1, number2){

        # Check that arguments provided are numberic
        if(class(number1) != "numeric" | class(number2) != "numeric"){
                print("Error: one or more of your inputs are not numeric")

        # Perform simple addition and return the result
        } else{
                result <- number1 + number2
                return(result)
```

```
        }
}
```

The function works

```
do_addition(3,3)
```

```
[1] 6
```

Forcada, Jaume, and Philip N. Trathan. 2009. "Penguin Responses to Climate Change in the Southern Ocean." *Global Change Biology* 15 (7): 1618–30. https://doi.org/10.1111/j.1365-2486.2009.01909.x.

Horst, Allison M., Alison Presmanes Hill, and Kristen B. Gorman. 2022. "The R Journal: Palmer Archipelago Penguins Data in the Palmerpenguins R Package - An Alternative to Anderson's Irises." *The R Journal* 14 (1): 244–54. https://doi.org/10.32614/RJ-2022-020.